

EQUIVARIANT QUANTUM NEURAL NETWORKS FOR IMAGE CLASSIFICATION: FROM MNIST TO SATELLITE IMAGERY

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Background: equivariance

A function f is *equivariant* under a symmetry group G if transforming the input by $g \in G$ and then applying f gives the same result as applying f first and transforming its output by a corresponding action of g : $f(g \cdot x) = g \cdot f(x)$.

For an image classifier this means: rotate or reflect the input, and the model’s internal representation transforms in a matched, predictable way, rather than being treated as an unrelated new input. Building this constraint directly into the circuit’s generators — instead of hoping data augmentation teaches it — is expected to need less data and generalize more reliably to transformed inputs never explicitly seen during training.

Motivation

Variational quantum circuits whose generators are built to commute with an image’s spatial symmetry group (**Equiv**) are compared against a matched, gate-for-gate identical circuit with that symmetry deliberately broken (**NonEquiv**). Both are trained on two binary 16×16 image-classification tasks: handwritten digits (MNIST) and a harder, real-world-like task — satellite imagery of ships vs. planes.

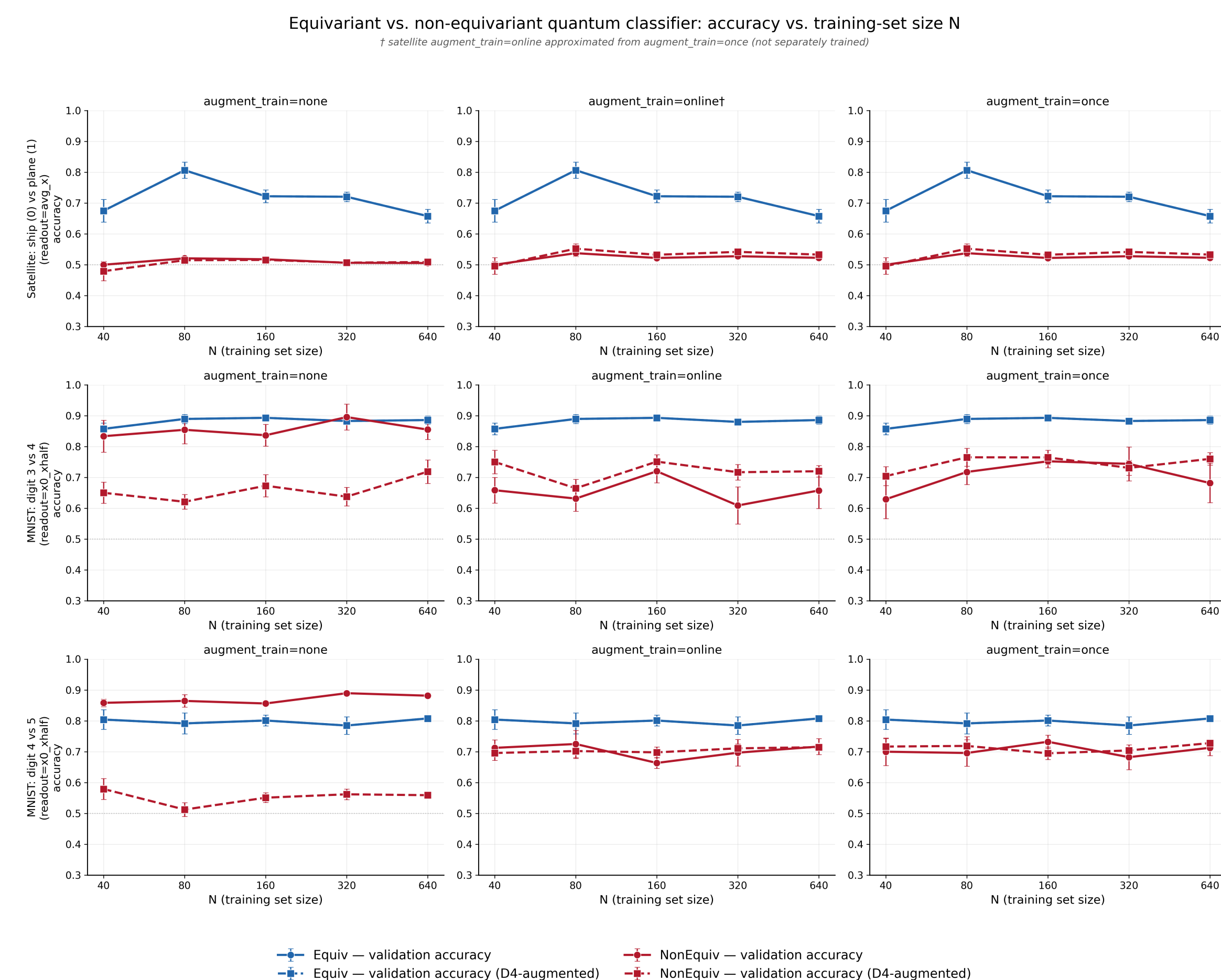
Method

Each 16×16 image is L2-normalized and amplitude-encoded onto 8 qubits (4 “row” + 4 “column” coordinate qubits), so the resulting quantum state’s amplitudes are exactly the image’s pixel values, only reindexed (encoding diagram: middle column).

Two matched 6-parameter ansätze (from the same published circuit family) are compared — identical gate count, wiring, and depth, differing only in which single-qubit rotations and which 4-qubit entangling gate are used (circuit diagrams: middle column). **Equiv**’s generators commute with the $p4m$ point group (row-flip, column-flip, row↔column swap); **NonEquiv** is an axis-scrambled variant with that symmetry deliberately broken.

Classification reads out either $\langle X \rangle$ averaged over all 8 qubits (**avg_x**) or $\langle X \rangle$ on 2 specific qubits (**x0_xhalf**) — the pair the ansatz’s own pooling schedule is designed to funnel information toward.

Results: accuracy vs. training-set size N

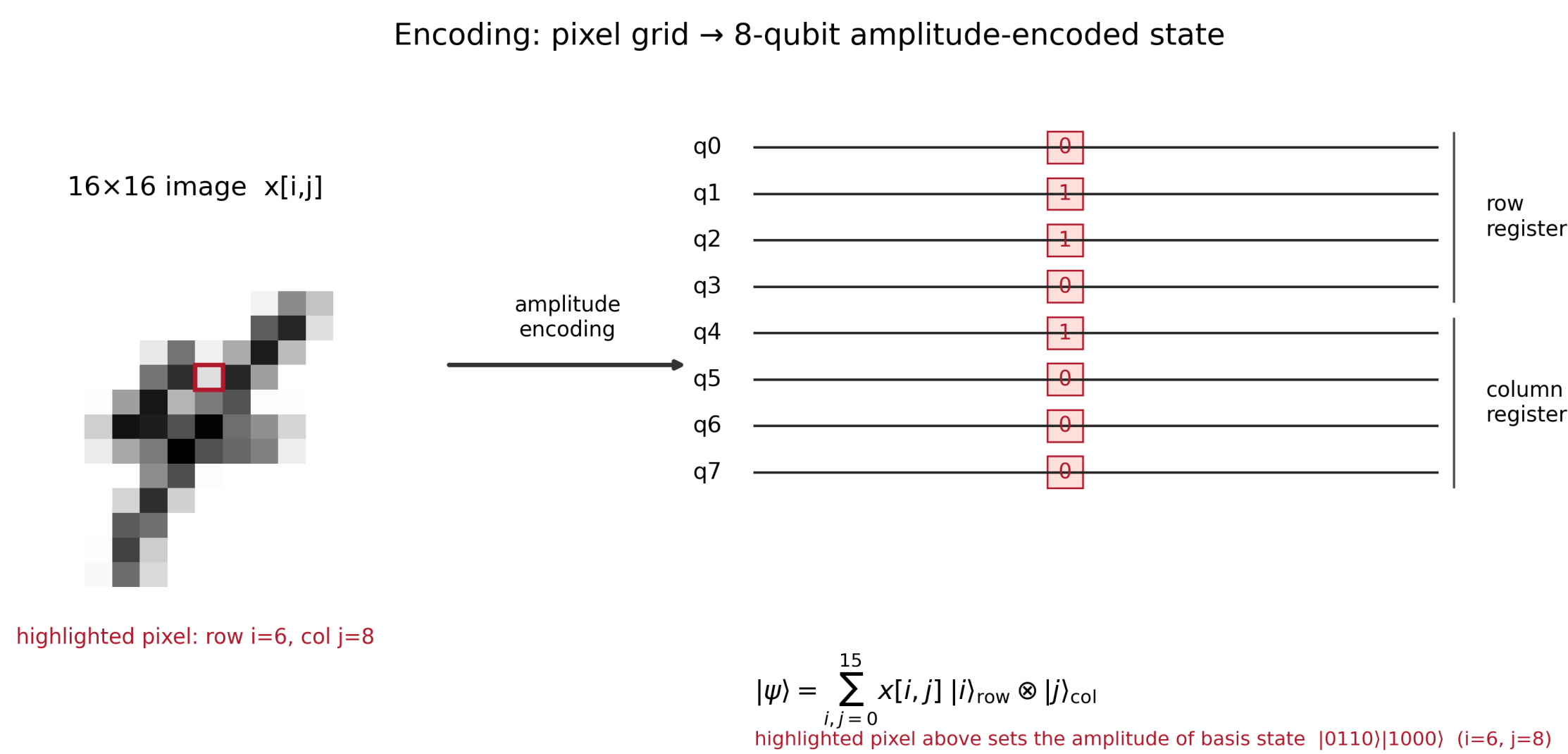


Mean $\pm$ SEM over 6 random seeds. Solid = clean validation accuracy; dashed = accuracy on the same images under a random D_4 (rotation/reflection) transform. † satellite **augment_train=online** was not separately trained — shown as its **once** run, which is expected to be extremely similar.

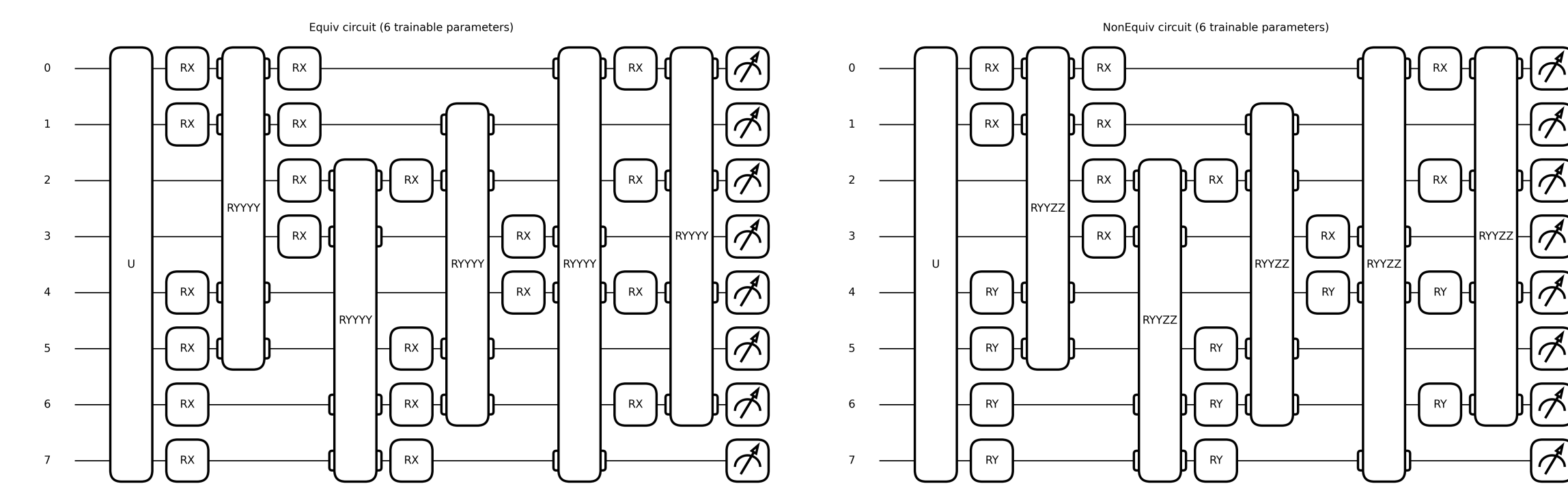
Conclusion

On a harder, real-world-like task, the practical bottleneck for this small (6-parameter) equivariant ansatz was not encoding fidelity — a classical model on the same preprocessed input recovers most of the discriminative signal — but training length and readout choice. Equivariance still delivered its core promise, better generalization under symmetry transformations of the input, even where raw accuracy on the harder dataset remained well below MNIST’s.

Encoding

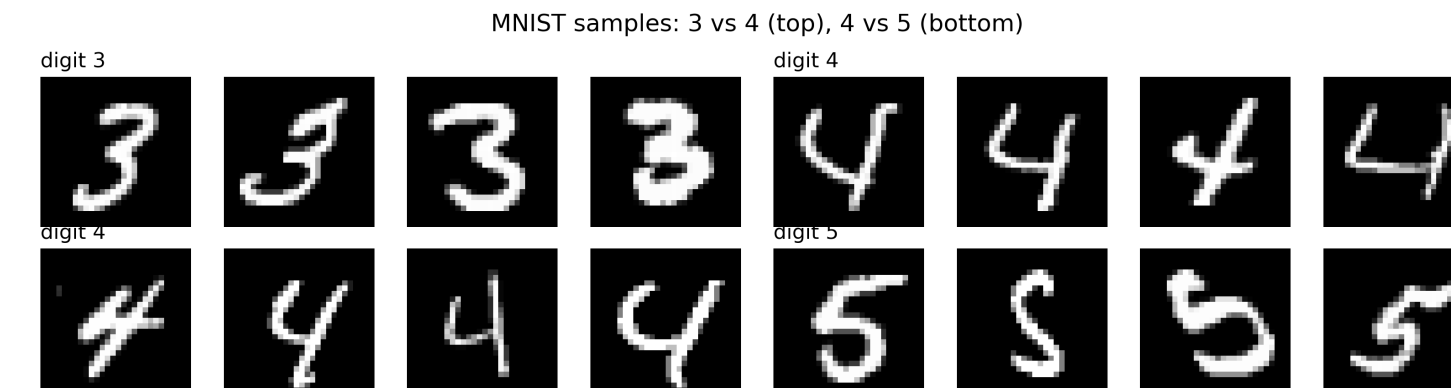


The two circuits (Equiv, left; NonEquiv, right)



Datasets

MNIST: two digit pairs (3 vs 4, 4 vs 5), standard train/test split.



Satellite: ship (rhammell/ships-in-satellite-imagery) vs. plane (rhammell/planesnet) chips, positive-class images only, resized to 16×16 grayscale.



Each class’s images are split once, deterministically, into train/test pools *before* any N -based sampling, then sampled $N/2$ -per-class from each pool — guaranteeing exact 50/50 balance in both train and test at every N (up to $N=1000$, bounded by the smaller ships pool of 1000 images).

Key findings

- **Equivariance buys robustness, not just raw accuracy.** Equiv’s accuracy on D_4 -transformed images tracks its clean accuracy closely on *both* datasets; NonEquiv’s drops sharply (e.g. MNIST 4 vs 5: 88% $\rightarrow$ 56% under **augment_train=none**).
- **The best readout is dataset-dependent.** **x0_xhalf** wins on MNIST ($\sim 89\%$); **avg_x** wins on satellite ($\sim 72\text{--}80\%$). A per-qubit $\langle X \rangle$ profiling experiment found the *same* qubits (0, 2, 4, 6) carry the strongest signal in both cases — MNIST’s signal is just $\sim 3\times$ stronger overall, so reading only 2 of those qubits is still enough there, but not on satellite.
- **Training duration mattered far more than expected** on the harder task. At 40 epochs, satellite accuracy averaged 65.5% (std 10.5% — two of six seeds still near chance); at 70 epochs it reached 74.8% (std 0.9%). The training *loss* looked flat from epoch ~ 10 while accuracy kept improving for another 60 epochs — a flat loss does not necessarily mean a flat decision boundary.
- **A classical linear baseline** (logistic regression on the same preprocessed pixels) reaches 80–99% on satellite, well above any quantum result found — a real, unclosed gap.
- **Naive attempts to close that gap underperformed.** Adding 2-qubit entangling gates, doubling or tripling circuit depth, and restricting the readout to only the empirically-strongest qubits all did *worse* than the simple **avg_x** baseline.